



Optimal pricing strategy for content products under competition: Pay-as-you-want or fixed-price?

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ABSTRACT

PAYW (pay as you want) has become a popular pricing strategy for content products in practice and has received considerable attention in the literature. However, the previous literature mainly justifies this pricing strategy in the monopoly setting. By exploiting a duopoly setting, this paper analytically investigates the impact of competition on the value of PAYW by comparing it to the traditional fixed pricing (FP) strategy. We find that only symmetric equilibria exist, i.e., both sellers choose PAYW when the fairness ideal is sufficiently high in equilibrium; otherwise, both choose FP. In sharp contrast to the prior literature, we find that multiple equilibria can exist, and a prisoner's dilemma situation can arise when the fairness ideal is at a medium level. We also find that the fairness ideal has a nonmonotonic impact on social welfare, and the presence of network externality makes PAYW preferable under competition. We also show that our results qualitative hold when consumers have heterogeneous product values.

1. Introduction

Pay-as-you-want (PAYW) is a consumer participative pricing mechanism in which consumers have full power to name the price of the product. This pricing strategy can be traced back to street shows. Street artists make their living by providing art performances in the street, and audience members can give (or not give) money freely depending on their preferences. In the era of E-Commerce, this pricing strategy has been widely adopted to sell content products in online marketplaces, such as e-readings, audio/video tracks, and online live streaming. The most successful example of the PAYW strategy for a content product may be the band Radiohead. This band announced in October 2007 that they would adopt the PAYW pricing strategy for their new album "In Rainbows". This album made \$3 million under the PAYW pricing strategy and created much social buzz. The data shows that consumers who download this album pay an average of \$2.26, which is more than that with a label company (\$2.25) or iTunes (\$1.40). Similar practices are also observed in museums, theatres, zoos, hotels, and restaurants (Chen et al., 2017).

The popularity of PAYW also attracts considerable attention in the literature. Empirical and experimental evidence suggests that some behavioral factors affect the feasibility of PAYW (e.g., Kim et al., 2009,

Gneezy et al., 2010, Kunter 2015, Weisstein et al., 2016, Barone et al., 2017). Fairness concern and reciprocity are the two most significant factors. Consumers with fairness concerns care about the relative surplus between themselves and others (Fehr and Schmidt 1999). Gneezy et al. (2012) showed that fairness concern leads consumers to pay a "fair price" under the PAYW strategy. Reciprocity refers to a behavioral response to an action perceived as either kind or unkind. (Falk and Fischbacher, 2006). Regner (2015) pointed out that reciprocity is the primary driver for a consumer's generous payments by using the Mag-nature data, while the reciprocity benefit is not present under a fixed pricing strategy (Machado and Sinha 2013). Considering these behavioral factors, theoretical studies mainly compare the profitability of PAYW with the traditional FP strategy to justify PAYW use, mainly in monopoly settings (e.g., Machado and Sinha, 2013, Isaac et al., 2015, Mak et al., 2015).

However, PAYW is also common in competitive environments (Krämer et al., 2017). For instance, influencers may post the same content, such as photos, records, and videos, on different social media platforms to allow fans from different platforms to tip. Some influencers also sell part of such content to different magazines, other influencers, or social media platforms to advertise their work. Unfortunately, the literature typically ignores the impact of competition, so it remains

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unclear how competition can affect profitability and social welfare under the PAYW strategy. In particular, it is still unclear how these behavioral factors affect sellers' pricing strategies in competitive environments. Furthermore, the network externality is a distinguishing feature for content products in competitive environments, e.g., consumers may benefit or suffer from other consumers consuming the same product (Shapiro et al., 1999). It is interesting to ask how network externality affects the value of PAYW, particularly under competition.

In this paper, we investigate the impact of competition on the value of PAYW. Specifically, we compare the PAYW strategy and the traditional FP strategy under duopoly competition. Similar to Chen et al. (2017), we build a two-stage behavioral game-theoretical model where consumers are fair-minded, and two sellers are located in a Hotelling line. Each seller chooses its pricing strategy, PAYW or FP, in the first stage, and they make price decisions in the second stage simultaneously. Different from Chen et al. (2017), consumers in our setting can gain added utility under PAYW, motivated by the empirical evidence that reciprocity is also a significant behavioral factor (Regner 2015). Another major difference from Chen et al. (2017) is that the transportation cost also affects the consumer's reference price, which is natural since the latter is a discount of the surplus in a transaction. This will be explained in detail in subsequent analysis.

We explicitly characterize the sellers' pricing strategy decisions and the equilibrium under competition. Also, we analyze the implications of the PAYW strategy on social welfare and the impact of network externality. Our main findings are summarized as follows.

1. Firstly, we find that only symmetric equilibria exist in our setting by modeling reciprocity. This result extends Chen et al. (2017) by introducing reciprocity. And we provide theoretical foundations for empirical evidence (Schmidt et al., 2015) about the significance of reciprocity to promote the use of the PAYW strategy.

2. Secondly, we find that the equilibrium may not be unique, and a prisoner's dilemma situation can arise in this case. This sharply contrasts with the previous literature, such as Chen et al. (2017), where the Nash equilibrium is unique. This finding encourages online platforms to adopt the PAYW pricing strategy when facing fierce competition.

3. Thirdly, we find that consumers' fairness ideal has a non-monotonic effect on social welfare. That is, social welfare will be optimal under a high consumers' fairness ideal. This reminds regulators and sellers to pay attention to the impact of consumers' fairness concerns when implementing the PAYW strategy in competitive environments.

4. In addition, we also investigate the impact of the network externality and the equilibrium when consumers are heterogeneous. The results show that the network externality can increase the profitability of PAYW and thus promote the application of PAYW under competition. And interestingly, we find that when there are sufficiently many low-value consumers in the market, asymmetric equilibrium may arise.

The rest of this article is organized as follows. In Section 2, we introduce the related literature. Section 3 presents the model and preliminary results. Section 4 analyzes the equilibria under competition. In Section 5, we discuss the implications on social welfare and the impact of network externality. We also investigate the impact of consumer heterogeneity in Section 5. Finally, we conclude in Section 6. Notation and all proofs are relegated to the Appendix.

2. Literature review

Our work directly contributes to the literature on the PAYW pricing strategy, particularly with competition. The experimental and empirical literature on PAYW is extensive. This stream of literature suggests that PAYW is suitable for content products with low marginal cost (Greiff and Egbert, 2018), and it identifies various factors that drive the wide application of PAYW, such as fairness concerns, price consciousness, reciprocity, social preference, loyalty and situational involvement (e.g., Kim et al., 2009, Gneezy et al., 2010, Gneezy et al., 2012, Barone et al., 2017, Krämer et al., 2017, Sharma et al., 2020, Roy and Das, 2021).

Among those factors, fairness concern and reciprocity are the two most significant (Kim et al., 2009). We refer readers to Natter and Kaufmann (2015) and Gerpott (2016) for recent comprehensive reviews.

Our work, however, joins the analytical studies on PAYW. This stream of literature mainly investigates the value of PAYW in a monopoly setting. For instance, Chao et al. (2015) identify conditions under which PAYW is more profitable than FP considering product marginal cost and consumer social preference. Chen et al. (2017) also compare PAYW with FP by considering consumers' fairness concern, showing that PAYW can help a seller to penetrate the market and enforce price discrimination. Other studies also find similar results from different perspectives (e.g., Isaac et al., 2015, Mak et al., 2015, Kahsay and Samahita, 2015). Unlike the above literature, we are now focusing on the impact of competition on the value of PAYW.

Studies on PAYW in a competitive environment are scarce. Chen et al. (2017) show that when consumers are sufficiently generous, both sellers in a duopoly competition prefer to choose PAYW and that PAYW can alleviate price competition. Considering social preference in PAYW, Chao et al. (2019) show that an asymmetric equilibrium with one seller choosing PAYW and the other choosing FP may arise to avoid the Bertrand trap. Chawan (2019) expands Chao et al. (2019) by considering free-loaders and warm-glow, and Samahita (2020) shows that entrants prefer PAYW over FP while the incumbent prefers FP. Similar to Chen et al. (2017), we also find that only symmetric equilibria exist in our setting. Unlike the above literature, we show that multiple equilibria can exist in our setting. As a result, a prisoner's dilemma situation can arise where both sellers choosing PAYW is dominated by both choosing FP. In addition, we also investigate the implications for social welfare and the impact of network externality in competition environments, which are also new topics in the literature.

Our model is closely related to Chen et al. (2017) in the sense that consumers are fair-minded, and the two sellers compete in a Hotelling line. However, there are two major differences in our model. First, we introduce and model consumers' reciprocity from PAYW, which is absent in Chen et al. (2017). Note that there is strong empirical evidence showing that reciprocity plays a critical role in consumers' payments and is a key driving factor for the adoption of PAYW (Schmidt et al., 2015, Regner 2015, Lu et al., 2021). Our results provide theoretical foundations for such empirical evidence. For instance, our symmetric equilibrium coincides with the laboratory experiments by Schmidt et al. (2015) showing that both sellers use PAYW when facing sufficiently altruistic consumers, and both use FP otherwise. Second, when modeling consumer's fairness concern under competition, we include the transportation cost into the consumer's "fair price". The transportation cost can be viewed as the hassle cost or disutility incurred by consumers when buying, accessing, or using the product, which is part of the total or net surplus of consumers. Since the fair price is the consumers' objective evaluation of the utility obtained from the product, it is reasonable to include transportation expenses in the consumers' fair price. As a result, we find that multiple equilibria and an interesting prisoner's dilemma situation can arise in our setting, though these results do not hold in Chen et al. (2017).

In summary, our work has three major contributions to the literature on PAYW strategy in competitive environments. First, we model reciprocity and transportation cost into the consumers' utility under PAYW and provide theoretical foundations for the related empirical evidence. Second, we expand the research field on PAYW strategy under competition. We obtain new insights into how consumers' fairness concern impacts the PAYW strategy with competition. In particular, we indicate that multiple equilibria and a prisoner's dilemma may arise. Third, we also investigate the management implications for social welfare, network externality, and heterogeneous consumers on the application of PAYW in competitive environments. These results and insights are also new in the literature and can enhance the management and application of PAYW strategy.

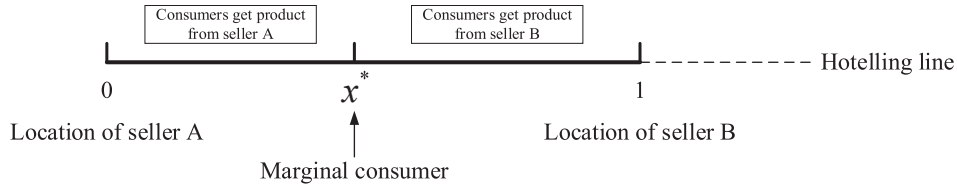


Fig. 1. Duopoly competition model.

3. The model

Consider a two-stage duopoly competition setting where two sellers sell the same content product. Following literature such as [Hitt and Chen \(2005\)](#), we assume that the marginal cost of the content product is zero. Most content products are information goods, such as videos, music, e-books, etc. The transmission and replication of content products almost require zero cost, compared with the traditional physical product. In the first stage, seller A and seller B decide their pricing strategy simultaneously. Each seller chooses between two pricing strategies: (1) the fixed pricing (FP) strategy, i.e., the seller directly decides the price for the product, and (2) the pay-as-you-want (PAYW) strategy, i.e., the seller does not decide the product price and lets consumers tip or pay for the product based on their own willingness. Hence, there are four subgames in the second stage. Once the pricing strategy is announced, consumers decide whether to buy the product and, if so, from which seller. Considering the assumption that seller A and seller B are symmetric, we only need to analyze the following three cases: a) both sellers choose to implement the FP strategy; b) both sellers choose to implement the PAYW strategy; c) one seller chooses FP while another seller chooses PAYW.

Following literature such as [Chen et al. \(2017\)](#), we assume that consumers are uniformly distributed in a classical Hotelling line between $[0, 1]$ with the sellers located at the ends of the line: seller A at $x = 0$, and seller B at $x = 1$. Consumers are homogenous about the value of the product, say r , but are heterogeneous in their locations, i.e., the consumer at location x incurs a transportation cost tx to buy from seller A and $t(1-x)$ from seller B. The transportation costs can be interpreted as the inconvenience cost or disutility incurred by consumers when buying, accessing, or using the product, or the consumers' different preferences for sellers. Fig. 1 depicts the classical Hotelling model described above.

Following the literature ([Haitao Cui et al., 2007](#) and [Chen et al., 2017](#)), consumers in our setting are fair-minded. Fair-minded consumers mean that consumers are inequity averse and prefer fair results. In other words, unfair results will have certain negative effects on fair-minded consumers' utility ([Fehr and Schmidt, 1999](#)). That is, consumers have an anchor price in mind, which we refer to as the "fair price", and they get an additional disutility if they pay higher or lower than this fair price. Note that fair price exists under both FP strategy and PAYW strategy. We assume each consumer's fair price is proportional to that consumer's net utility (e.g., [Caliskan-Demirag et al., 2010](#), [Yoshihara and Matsubayashi, 2021](#)). For example, suppose a consumer at location x buys from seller A. Then, we use $\lambda(r-tx)$ to denote the fair price. That is, consumers believe that the fair surplus obtained by seller A should be $\lambda(r-tx)$, and the fair surplus obtained by themselves should be $(1-\lambda)\lambda(r-tx)$. The parameter $\lambda \in (0, 1)$ is referred to as the consumer's fairness ideal, which is a common concept in the literature that captures how the fair allocation is formed ([Cui and Mallucci, 2016](#)). In our setting, it can be interpreted as the seller's share of the total surplus from the perspective of consumers.

Consumers' fairness concern can affect their utilities from advantageous unfairness concern and disadvantageous unfairness concern ([Fehr and Schmidt, 1999](#)). When consumers pay more than the fair price, disadvantageous unfairness concern will affect their utilities. That means consumers obtain disadvantageous results in unfair distributions.

Similarly, advantageous unfairness concern affects consumers' utilities when consumers pay less than the fair price. Following the literature ([Cui and Mallucci, 2016](#), [Chen et al., 2017](#), [Kang et al., 2021](#)), we also assume the disutility from these two types of fairness concerns is linear. Let β measure the degree of the disadvantageous unfairness concern and μ measure the degree of the advantageous unfairness concern. Then, for a consumer at location x buying from seller A, the disutility from advantageous unfairness concern is $-\beta \max\{p_A - \lambda(r-tx), 0\}$, and the disutility from disadvantageous unfairness concern is $-\mu \max\{\lambda(r-tx) - p_A, 0\}$. We assume $\beta > \mu$, which is consistent with the empirical evidence that disadvantageous unfairness has a stronger impact on consumer utility than advantageous unfairness ([Bolton and Ockenfels, 2000](#), [Loch and Wu, 2008](#)). Without loss of generality, we assume $0 < \mu < 1$.

We next consider the consumer's utility function under different pricing strategies. For the FP strategy, we assume p_{FP} denotes the fixed price set by the seller. Then, the utility of the consumer at location x under the FP strategy can be formulated as follows:

$$u_{FP} = r - p_{FP} - \beta \max\{p_{FP} - \lambda(r-tx), 0\} - \mu \max\{\lambda(r-tx) - p_{FP}, 0\} - tx \quad (1)$$

Different from the FP strategy, consumers can gain additional positive utility due to reciprocity under PAYW ([Rankin and Eggimann, 2009](#), [Machado and Sinha, 2013](#), [Kim et al., 2014](#)). To capture the effect of reciprocity, we assume the additional utility caused by reciprocity positively relates to the price, following [Machado and Sinha \(2013\)](#). Specifically, let $\alpha \in [0, 1]$ denote the sensitivity to reciprocity and p_{PAYW} denote the price paid by consumers under PAYW. Then αp_{PAYW} represents the additional utility gained by the consumer under PAYW. The utility for consumers at location x under PAYW strategy can be formulated as follows:

$$u_{PAYW} = r - p_{PAYW} + \alpha p_{PAYW} - \beta \max\{p_{PAYW} - \lambda(r-tx), 0\} - \mu \max\{\lambda(r-tx) - p_{PAYW}, 0\} - tx \quad (2)$$

Note that consumers pay for the product based on their own willingness under PAYW. In other words, consumers under PAYW decide on p_{PAYW} to maximize their total utility u_{PAYW} . The following lemma presents consumer's strategic decision under PAYW.

Lemma 1. Consumer's payment and utility under PAYW are as follows:

(a) When $\alpha + \mu < 1$, the consumer's optimal payment and net utility are $p_{A_PAYW}^* = p_{B_PAYW}^* = 0$, $u_A = (r-tx)(1-\mu\lambda)$, $u_B = [r-t(1-x)](1-\mu\lambda)$, respectively.

(b) When $\alpha + \mu \geq 1$, the consumer's optimal payment and net utility are $p_{A_PAYW}^* = \lambda(r-tx)$, $p_{B_PAYW}^* = \lambda[r-t(1-x)]$, $u_A = (r-tx)(1-\lambda+\alpha\lambda)$, $u_B = [r-t(1-x)](1-\lambda+\alpha\lambda)$, respectively.

Lemma 1 distinguishes two cases. For case (a), consumers can enjoy the product and pay nothing. Consumers doing this are free-riding, so PAYW can never generate positive profit for the seller in this situation. Since we are interested in the value of PAYW under competition, we assume $\alpha + \mu \geq 1$ holds in what follows to rule out free-riding consumers. Thus, it is worth mentioning that consumers will pay the same price as the fair price for the content product.

Next, we present the consumers' and sellers' strategic decisions in

each subgame.

3.1. Subgame (FP, FP)

In this subgame, both sellers choose the FP strategy and decide on a fixed price. Let p_A and p_B denote the price offered by sellers A and B, respectively. A consumer at location x buying from seller A or B can get surpluses as follows:

$$u_A = r - p_A - \beta \max\{p_A - \lambda(r - tx), 0\} - \mu \max\{\lambda(r - tx) - p_A, 0\} - tx$$

$$u_B = r - p_B - \beta \max\{p_B - \lambda[r - t(1 - x)], 0\} - \mu \max\{\lambda[r - t(1 - x)] - p_B, 0\} - t(1 - x).$$

Let x^* denote the consumer who is indifferent between buying from seller A or B.

It is straightforward to see those consumers at $x < x^*$ buy from seller A, and otherwise buy from seller B. Thus, the demand for seller A is x^* and the demand for seller B is $1 - x^*$.

Therefore, the sellers' problems can be formulated as follows:

$$\pi_A = \max_{p_A} p_A \times x^* - c$$

$$\pi_B = \max_{p_B} p_B \times (1 - x^*) - c$$

3.2. Subgame (PAYW, PAYW)

In this subgame, neither seller makes a strategic pricing decision. Instead, the consumers decide how much to pay and whether to buy from seller A or B. With a little abuse of notation, let p_A and p_B denote the "price" offered by a consumer at location x when buying from sellers A and B, respectively. The consumer's problem when buying from sellers A and B can be formulated as follows:

$$\max_{p_A} u_A = r - p_A + \alpha p_A - \beta \max\{p_A - \lambda(r - tx), 0\} - \mu \max\{\lambda(r - tx) - p_A, 0\} - tx$$

$$\max_{p_B} u_B = r - p_B + \alpha p_B - \beta \max\{p_B - \lambda[r - t(1 - x)], 0\} - \mu \max\{\lambda[r - t(1 - x)] - p_B, 0\} - t(1 - x).$$

Again, let x^* denote the consumer who is indifferent between buying from seller A or B. Then, the sellers' problems can be formulated as follows:

$$\pi_A = \int_0^{x^*} \max_{p_A} p_A dx - c = \int_0^{x^*} \lambda(r - tx) dx - c$$

$$\pi_B = \int_{x^*}^1 \max_{p_B} p_B dx - c = \int_{x^*}^1 \lambda[r - t(1 - x)] dx - c$$

3.3. Subgame (FP, PAYW)/subgame (PAYW, FP)

Since sellers A and B are symmetric, subgame (FP, PAYW) is essentially the same as subgame (PAYW, FP) by exchanging the roles of sellers A and B. Therefore, we only analyze subgame (PAYW, FP), i.e., seller A chooses PAYW while seller B chooses FP.

In this subgame, the consumer utility for those at location x buying from A and B are as follows:

$$\max_{p_A} u_A = r - p_A + \alpha p_A - \beta \max\{p_A - \lambda(r - tx), 0\} - \mu \max\{\lambda(r - tx) - p_A, 0\} - tx,$$

$$u_B = r - p_B - \beta \max\{p_B - \lambda[r - t(1 - x)], 0\} - \mu \max\{\lambda[r - t(1 - x)] - p_B, 0\} - t(1 - x),$$

where p_B denotes the fixed price decided by seller B, and the consumer decides p_A to maximize their utility.

Seller A and B's profit can be formulated as follows:

$$\pi_A = \int_0^{x^*} \max_{p_A} p_A dx - c = \int_0^{x^*} \lambda(r - tx) dx - c$$

$$\pi_B = \max_{p_B} p_B \times (1 - x^*) - c.$$

4. Analysis

In this section, we first analyze the sellers' optimal decision under each subgame, then we compare the results in all subgames to obtain the equilibrium.

4.1. Subgame analysis

4.1.1. (FP, FP) subgame

In this section, we analyze the case where both sellers adopt the FP strategy.

Lemma 2. When both sellers adopt the FP strategy, the optimal fixed price of each seller is $\frac{t(1-\mu\lambda)}{1-\mu}$ and the profit is $\frac{t(1-\mu\lambda)}{2(1-\mu)} - c$.

Since the setting is symmetric, both sellers set the same fixed price and earn the same profit, and each covers half of the market. Furthermore, the optimal price and profit increase in the unit transportation cost t and decrease in the fairness ideal λ . In addition, the FP seller will set the fixed price higher than the fair price, which will make consumers suffer from advantageous unfairness. Also notice that when $t = 0$, both sellers earn zero profit and fall into the Bertrand trap.

4.1.2. (PAYW, PAYW) subgame

In this section, we analyze the case where both sellers adopt the PAYW strategy.

Lemma 3. When both sellers adopt the PAYW strategy, their profit is $\frac{\lambda}{2}(r - \frac{t}{4}) - c$. The consumers' payment for seller A's content product is $\lambda(r - tx)$, and the payment for seller B's content product is $\lambda[r - t(1 - x)]$.

Again, since both sellers adopt the PAYW strategy, they earn the same profit, and each seller sells a content product to half of all consumers. Based on the analysis in Lemma 1, the consumer's payment for the content product is the same as the fair price. And this payment increases with λ and decreases with t . In addition, the seller's profit is decreasing with the unit transportation cost t . This lemma also hints that it is profitable for both sellers to adopt PAYW when their content products have little transportation cost and low fixed cost.

4.1.3. (PAYW, FP) subgame

In this section, we analyze the case where one seller adopts PAYW, and the other adopts FP. Without loss of generality, we only analyze the case that seller A chooses PAYW and seller B chooses FP since the other case (FP, PAYW) can be treated similarly.

Lemma 4. When seller A adopts the PAYW strategy and seller B adopts the FP strategy in a competitive market, the consumer located at $x^* = \frac{3}{4}$ is indifferent about getting the content product from seller A or seller B. The profits of sellers A and B can be expressed as $\pi_A = \frac{3}{4}\lambda(r - \frac{3}{8}t) - c$ and $\pi_B = \frac{t(1-\mu\lambda)}{8(1-\mu)} - c$. The optimal fixed price of seller B is $p_B^* = \frac{t(1-\mu\lambda)}{2(1-\mu)}$.

Different from the other two symmetric cases, when sellers adopt different pricing strategies, the PAYW seller occupies a higher share of the market ($x^* = \frac{3}{4}$). Said differently, PAYW brings more consumers to sellers in a competitive environment. Another interesting result is that the unit transportation cost t has different impacts on the sellers' profits. Specifically, the profit of the FP seller increases with t while the PAYW seller's profit decreases with t . Therefore, sellers in a more competitive market (a smaller t) prefer the PAYW strategy over the FP strategy.

Table 1
Profits of sellers in different subgames.

		B	
		FP	PAYW
A	FP	$\frac{t(1-\mu\lambda)}{2(1-\mu)} - c, \frac{t(1-\mu\lambda)}{2(1-\mu)} - c$	$\frac{t(1-\mu\lambda)}{8(1-\mu)} - c, \frac{3}{4}\lambda\left(r - \frac{3}{8}t\right) - c$
	PAYW	$\frac{3}{4}\lambda\left(r - \frac{3}{8}t\right) - c, \frac{t(1-\mu\lambda)}{8(1-\mu)} - c$	$\frac{\lambda}{2}\left(r - \frac{t}{4}\right) - c, \frac{\lambda}{2}\left(r - \frac{t}{4}\right) - c$

Comparing the profits of the above two sellers, we can see that when $\lambda < \frac{4t}{24r(1-\mu)+13t\mu-9t}$ the seller adopting FP earns more profit than its opponent adopting PAYW. From this condition, we can see that the PAYW strategy is less attractive when the consumers' fairness idea is low. The intuition is as follows. A low fairness ideal implies that consumers prefer to leave the seller only a small share of the transaction surplus, so they pay less under PAYW, which reduces the profitability of the PAYW seller.

4.2. Equilibrium pricing strategy

In this section, we analyze the equilibrium pricing strategy by comparing the above section's results in each subgame. Table 1 summarizes the sellers' profits in the subgames. The following proposition presents the equilibrium for the optimal pricing strategy.

Proposition 1. *The equilibrium under duopoly competition is as follows:*

- (a) If the fairness ideal is high, i.e., $\lambda > \lambda_3$, the unique equilibrium is that both sellers adopt the PAYW strategy.
- (b) If the fairness ideal is low, i.e., $\lambda < \lambda_2$, the unique equilibrium is that both sellers adopt the FP strategy.
- (c) If the fairness ideal is moderate (a medium level), i.e., $\lambda_2 < \lambda < \lambda_3$, then there are two symmetric equilibria when $\lambda_2 < \lambda < \lambda_1$ (i.e., either both sellers choose PAYW or both sellers choose FP), and one equilibrium but a prisoner's dilemma when $\lambda_1 < \lambda < \lambda_3$ (i.e., both sellers choose PAYW).

Furthermore, both sellers prefer the PAYW strategy if the market is highly competitive (t is small). The expressions for λ_1 , λ_2 , and λ_3 are given in the Appendix.

Proposition 1 provides insights into how two competing sellers choose an optimal pricing strategy to achieve equilibrium. Fig. 2 depicts the equilibrium outcomes. Interestingly, Proposition 1 suggests that only symmetric equilibrium can be sustained, i.e., sellers choose the same pricing strategy in equilibrium. This result contradicts Chao et al. (2019), who show that the asymmetric equilibrium (PAYW, FP) can arise and break the Bertrand trap. The reason behind this contrast is that Chao et al. (2019) ignore the consumer's fairness concerns in the FP strategy. This is not the case in our setting. In this sense, our setting and results are more consistent with Chen et al. (2017). However, different from Chen et al. (2017), Proposition 1 suggests that there could be multiple equilibria, and a prisoner's dilemma situation can arise when the fairness ideal is medium (i.e., when $\lambda_2 < \lambda < \lambda_3$). In other words, both sellers choosing FP can bring higher profits than both sellers

choosing PAYW in this situation. These findings also support the experimental results in the competitive environment of Schmidt et al. (2015). Note that the previous literature, including Chen et al. (2017), often suggests that (PAYW, PAYW) is the unique equilibrium for a large fairness ideal, and (FP, FP) is the unique equilibrium otherwise.

The rationale is as follows. From the perspective of profit, the market share advantage brought by PAYW cannot make up for the price disadvantage under a smaller λ . As λ increases, the price paid by PAYW consumers increases, giving the seller a profit and market share incentive to change from FP to PAYW. The profit and market share of the seller who does not change their strategy will be greatly reduced. Due to the setting of independent and symmetrical competing sellers, a prisoner's dilemma arises at this time. Note, however, that this situation benefits social welfare, as discussed in the next section. When λ increases to λ_3 , the fair price of consumers will become higher, and the average price consumers pay under (PAYW, PAYW) is higher than the fixed price set by the seller under (FP, FP). The prisoner's dilemma no longer exists at this time.

It is also worth mentioning that there are two equilibrium outcomes when the consumer's fairness ideal is between λ_2 and λ_1 . If the sellers have complete information while making decisions, a rational seller will always choose FP. However, if the seller has other considerations, it is also possible for the seller to choose PAYW. It is easy to prove that when one seller chooses PAYW and the other chooses FP, the PAYW seller will always get a higher market share and profit than the FP seller. In this case, the FP seller also has incentives to switch to PAYW. Unless both sellers firmly choose the FP strategy because of higher profit when $\lambda_2 < \lambda < \lambda_1$, the PAYW equilibrium is likely to occur.

Finally, Proposition 1 indicates that competition makes sellers more likely to choose PAYW. It can be seen from Table 1 that higher competition will increase the PAYW seller's profit while reducing the FP seller's profit. The values of λ_1 , λ_2 , and λ_3 will also decrease as competition intensifies (a smaller t), which makes a PAYW equilibrium more likely to occur. Under the FP equilibrium, sellers fall into the Bertrand competition trap as the competition intensifies, and each has the incentive to change from FP to PAYW. Under the PAYW equilibrium, consumers flexibly adjust their payment, and that alleviates the price competition.

5. Extensions

5.1. Welfare comparison

In this section, we investigate the impact of competition on social welfare. In the previous section, we obtained the sellers' profits. We next calculate consumer surplus and social welfare. Consumer surplus is defined as the total utilities obtained by consumers (Zhu et al., 2019, He et al., 2020), which can be expressed mathematically as $CS = \int_0^{x^*} u_A dx + \int_{x^*}^1 u_B dx$. It is worth mentioning that x^* represents the position of marginal consumers. Social welfare measures the benefits obtained by sellers and consumers (Hendershott and Zhang 2006, Kong et al., 2022), which can be expressed mathematically as $SW = CS + \pi_A + \pi_B$. Table 2

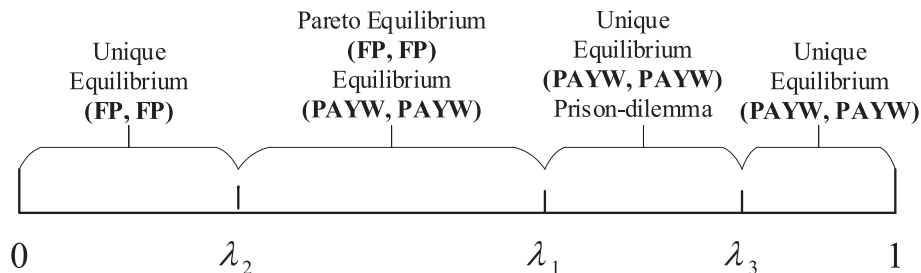


Fig. 2. The equilibrium outcomes.

Table 2
Consumer surplus in different subgames.

		B	
		FP	PAYW
A	FP	$\frac{(4r-5t)(1-\mu\lambda)}{8}$	$\frac{(8r-5t)(1-\mu\lambda)}{32}$
		$\frac{(4r-5t)(1-\mu\lambda)}{8}$	$\frac{(24r-9t)(1-\lambda+\alpha\lambda)}{32}$
	PAYW	$\frac{(24r-9t)(1-\lambda+\alpha\lambda)}{32}$	$\frac{(4r-t)(1-\lambda+\alpha\lambda)}{8}$
		$\frac{(8r-5t)(1-\mu\lambda)}{32}$	$\frac{(4r-t)(1-\lambda+\alpha\lambda)}{8}$

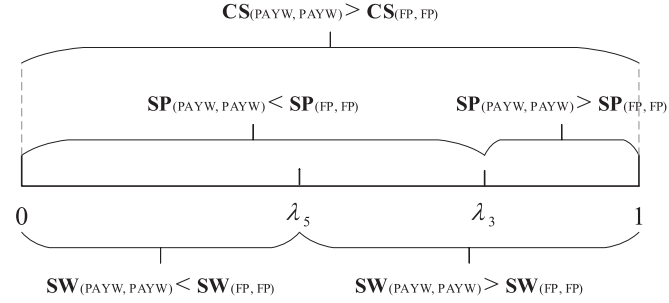


Fig. 3. Consumer surplus and social welfare under different λ under equilibrium.

shows the consumer surplus in the four subgames.

Proposition 2. Although larger λ brings higher profits to the PAYW seller, it lowers the consumer surplus. The consumer surplus under the PAYW equilibrium is higher than under the FP equilibrium ($\lambda_4 < 0$). Only when $\lambda_3 < \lambda$ is the entire market Pareto-optimal from the perspective of both sellers and consumers. The expressions for λ_4 and λ_5 are given in the Appendix.

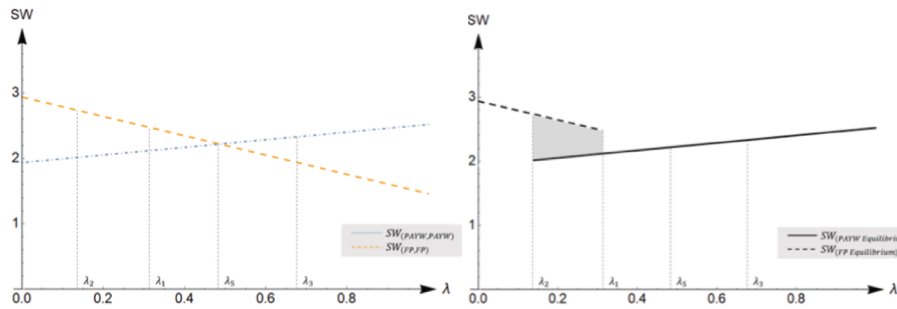
Proposition 2 shows that a relatively high fairness ideal is bad for consumers because it will hurt consumer surplus. Two interesting

observations arise from **Proposition 2**. First, the consumer surplus under the PAYW equilibrium will always be higher than under the FP equilibrium. The intuition behind this is that consumers are affected by fairness concern under FP, which will result in surplus losses. However, due to the fair price paid by consumers under the PAYW equilibrium, the consumer surplus under PAYW is not affected by fairness concern. **Chao et al. (2015)** and **Tudón (2015)** pointed out that consumer welfare is higher under PWYW strategy than that under FP strategy. Different from their monopoly setting, we expanded this research in a competitive environment. We indicate that PAYW seller always brings higher consumer surplus in a competitive environment, regardless of the pricing strategy adopted by another seller. Second, when $\lambda_3 < \lambda$, the entire market is at a Pareto optimum. In this case, both the consumer surplus and the seller's profit are higher under the PAYW equilibrium than under the FP equilibrium. That is, there is a win-win situation where sellers and consumers both get higher benefits in the PAYW equilibrium. This finding provides valuable implications for government or platform regulators, who could design tools to induce this Pareto optimum to benefit both sellers and consumers.

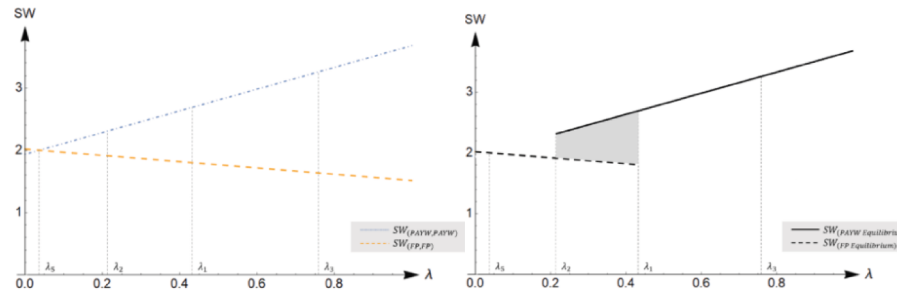
Proposition 3. A small $\alpha < \alpha_1$ causes social welfare loss in the PAYW equilibrium when facing a medium λ such that $\lambda_2 < \lambda < \lambda_1$. A large $\alpha > \alpha_2$

Table 3
Consumer surplus in subgames with network externality.

		B	
		FP	PAYW
A	FP	$\frac{(4r-5t+\frac{1}{2}\gamma)(1-\mu\lambda)}{8}$	$\frac{(8r-5t+6\gamma)(1-\mu\lambda)}{32}$
		$\frac{(4r-5t+\frac{1}{2}\gamma)(1-\mu\lambda)}{8}$	$\frac{(24r+18\gamma-3t)(1-\lambda+\alpha\lambda)}{32}$
	PAYW	$\frac{(24r+18\gamma-3t)(1-\lambda+\alpha\lambda)}{32}$	$\frac{(4r+2\gamma-t)(1-\lambda+\alpha\lambda)}{8}$
		$\frac{(8r-5t+6\gamma)(1-\mu\lambda)}{32}$	$\frac{(4r+2\gamma-t)(1-\lambda+\alpha\lambda)}{8}$



(a) Small α ($t = 0.5$ $r = 4$ $\alpha = 0.3$ $\mu = 0.8$)



(b) Large α ($t = 0.5$ $r = 4$ $\alpha = 0.9$ $\mu = 0.25$)

Fig. 4. Social welfare under different pricing strategies and different equilibrium.

Table 4
Profits of sellers in subgames with network externality.

		B	
		FP	PAYW
A	FP	$\frac{t(1-\mu\lambda)}{2(1-\mu)} - c, \frac{t(1-\mu\lambda)}{2(1-\mu)} - c$	$\frac{(t-\gamma)(1-\mu\lambda)}{8(1-\mu)} - c,$ $\frac{3}{4}\lambda\left(r - \frac{3}{8}t + \frac{3}{4}\gamma\right) - c$
	PAYW	$\frac{3}{4}\lambda\left(r - \frac{3}{8}t + \frac{3}{4}\gamma\right) - c,$ $\frac{(t-\gamma)(1-\mu\lambda)}{8(1-\mu)} - c$	$\frac{\lambda}{2}\left(r - \frac{t}{4} + \frac{1}{2}\gamma\right) - c,$ $\frac{\lambda}{2}\left(r - \frac{t}{4} + \frac{1}{2}\gamma\right) - c$

causes social welfare loss in the FP equilibrium. The expressions for α_1 and α_2 are given in the Appendix.

Proposition 3 states the influence of reciprocity on social welfare under equilibrium. Specifically, when faced with a small α (i.e., $\alpha < \alpha_1$), we have $\lambda_1 < \lambda_5 < \lambda_3$. In this case, for a medium-level fairness ideal (i.e., when $\lambda_2 < \lambda < \lambda_1$), the sellers' choice of the PAYW equilibrium lowers the social welfare because the social welfare under the FP equilibrium is greater than that under the PAYW equilibrium. For a large α (e.g., $\alpha > \alpha_2$), we have $\lambda_5 < \lambda_2 < \lambda_1$. Similarly, we can also show that the FP equilibrium gives lower social welfare. The results of our theoretical analysis support the empirical analysis conclusion of [Machado and Sinha \(2013\)](#) that PAYW strategy benefits from consumers' reciprocity. This is

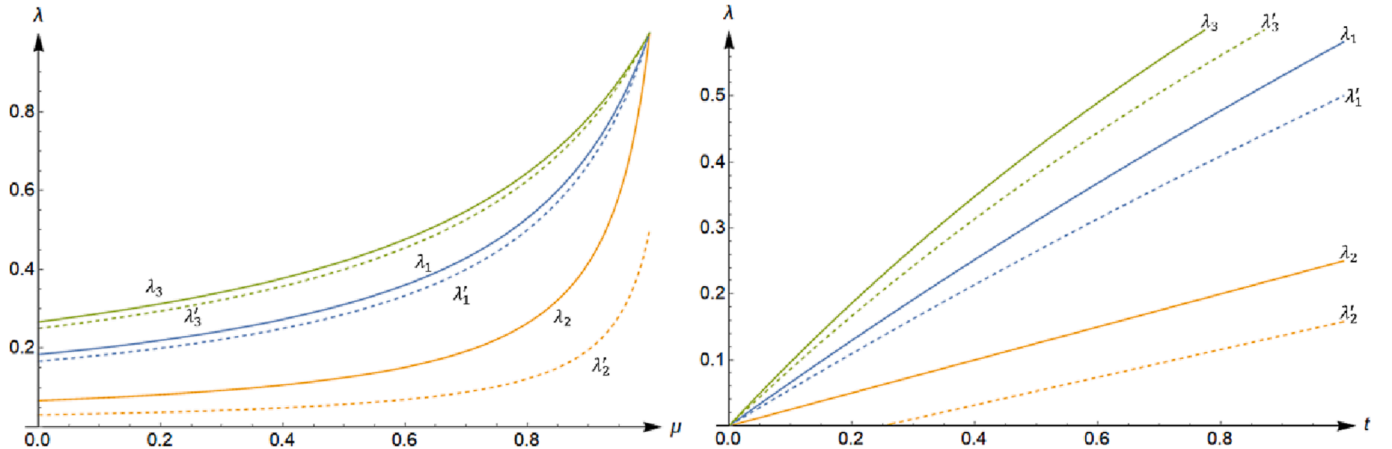


Fig. 5. Different λ' changes with μ (left) and t (right) ($\gamma = 0.5$).

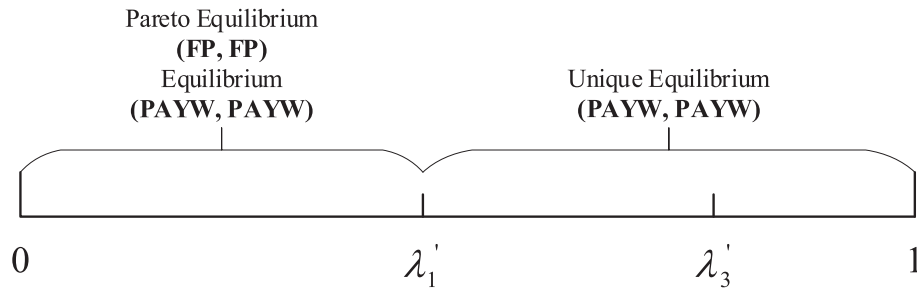


Fig. 6. Equilibrium outcomes when $t < \gamma$.

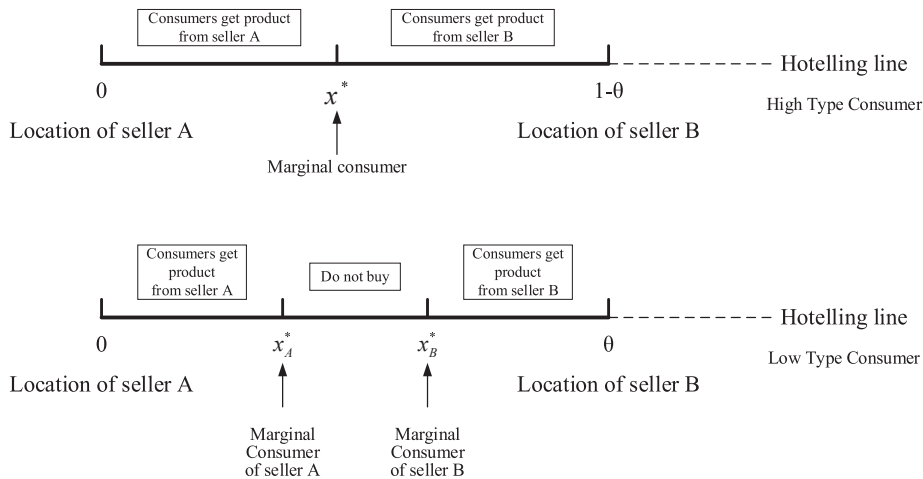


Fig. 7. Market coverage for the two types of consumers.

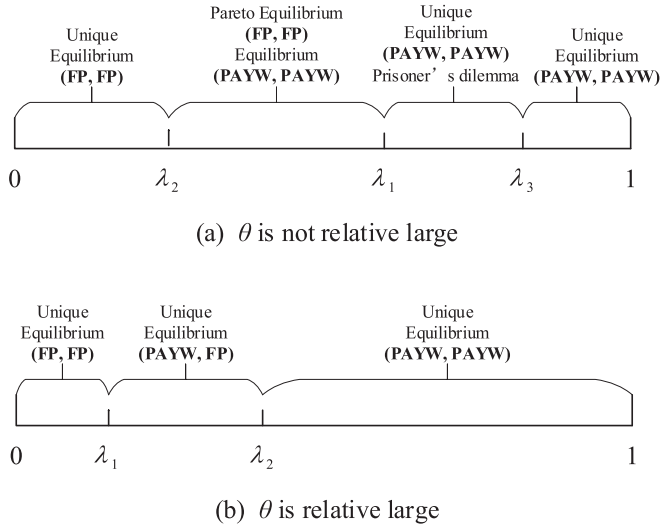


Fig. 8. The equilibrium outcomes with two types consumers.

Table 5
Model Notation.

r	Consumer's willingness to pay for the content product	
p_{FP}	Price of content product set by seller under the fixed price strategy	Decision variable Decision maker is seller
p_{PAYW}	Price voluntarily paid for the content product by the consumer under the PAYW strategy	Decision variable Decision maker is consumer
u	Consumer's utility in different strategies (denoted by the subscript FP or PAYW) in basic model or for different sellers (denoted by the subscript A or B) in competition model	
β	Degree of consumer's disadvantageous unfairness concern	
μ	Degree of consumer's advantageous unfairness concern	
α	Consumer's sensitivity to reciprocity	
λ	Consumer's fairness ideal (seller's fair share of the total surplus)	
π	Seller's profit in different strategies (denoted by the subscript FP or PAYW) in basic model or for different sellers (denoted by the subscript A or B) in competition model	
c	Fixed cost of content product	
θ	Proportion of consumers who pay zero under PAYW strategy	
t	Unit transportation cost in Hotelling competition model	
γ	Strength of the network externality	

because reciprocity leads to a higher surplus for the PAYW consumers and no additional benefit for the FP consumers.

Propositions 2 and 3 together suggest that the consumer's fairness ideal level has a nonmonotonic effect on social welfare. Counterintuitively, a medium consumer's fairness ideal level may bring the lowest social welfare. To show this finding more intuitively, consider Figs. 3 and 4. Fig. 3 shows the situation of social welfare, seller profit, and consumer surplus under different λ under the equilibrium outcomes. In the figure, we use CS to represent consumer surplus, SP to represent the seller's profit, and SW to represent social welfare. The subscripts indicate different equilibrium strategies. Fig. 4 shows how social welfare varies under different equilibria. We can see that the social welfare under the PAYW equilibrium increase with λ , while the social welfare decreases with λ under the FP equilibrium. The possible loss of social welfare can also be seen in this figure, indicated by the shaded part in

Fig. 4.

5.2. Network externalities

In this section, we investigate how our insights vary in the presence of network externality, which is a common feature for content products (e.g., Griva and Vettas 2011, Cheng and Tang 2010, Zhang and Seidmann 2010, Yang and Mai 2010). As previously mentioned, network externality refers to the consumers benefiting from or getting hurt by other consumers consuming the same product (Shapiro et al., 1999). We adopt the model in Cheng and Tang (2010) to capture network externality by saying it enlarges the utility of consumers by γQ . The parameter γ represents the strength of the network externality, and Q represents the quantity (number) of consumers. Therefore, when a consumer at location x buys the content product from seller A under FP, their utility is $u_A = r - p_A - \beta \max\{p_A - \lambda(r - tx), 0\} - \mu \max\{\lambda(r - tx) - p_A, 0\} - tx + \gamma Q_A$. The same consumer, buying from seller A under PAYW strategy, has utility $u_A = r - p_A + \alpha p_A - \beta \max\{p_A - \lambda(r - tx), 0\} - \mu \max\{\lambda(r - tx) - p_A, 0\} - tx + \gamma Q_A$. Similarly, we can get the utility expressions when consumers choose seller B. Using the other assumptions above, we normalize the quantity of consumers in the market to 1 and assume the reserve price of consumers in the market is sufficient to cover the market.

Next, we analyze the equilibrium outcomes in the duopoly market while focusing on the influence of network externalities on the equilibria. Tables 3 and 4 present the consumer surplus and sellers' profits in different subgames with network externality.

We have the following findings in the above tables. First, when both sellers adopt FP, network externalities will not affect corporate profits but will significantly increase the consumer surplus. Second, when both sellers adopt PAYW, the existence of network externalities significantly impacts corporate profits and significantly increases the consumer surplus. Third, when seller A adopts PAYW and seller B adopts FP, the existence of network externalities does not affect the seller's market share, but those externalities significantly increase seller A's profits and significantly reduce seller B's profits. At the same time, the existence of network externalities increases the consumer surplus of seller A and seller B. Therefore, we have the following proposition about the impact of network externality on the PAYW strategy.

Proposition 4. *The presence of network externalities makes the PAYW strategy more preferable in a competitive environment for the seller.*

Proposition 4 states that network externality promotes PAYW in competitive environments. This is consistent with practice, as some platforms add social functions (such as comments in a music platform and a bullet screen in a video platform) when selling content products. To understand the impact of network externality, one can see from Fig. 5 that the threshold of the equilibrium will move to the left when network externalities exist (i.e., $\lambda'_i < \lambda_i$, where $i = 1, 2, 3$). The expressions for λ'_1 , λ'_2 , and λ'_3 are given in the Appendix. This implies that the presence of network externalities makes PAYW more profitable, and so the PAYW equilibrium is more likely to occur. Our findings are different from the results obtained in previous literature on the impact of network externalities on PAYW strategy. For example, El Harbi et al. (2014) indicated that PAYW strategy is more profitable than FP strategy in the presence of network externalities, but Kim et al. (2022) argued that the FP seller benefits more from network externalities compared with the PAYW seller. These conclusions are all found in a monopoly environment, while we are the first to explore the impact of network externalities on the application of the PAYW strategy in a competitive environment. The intuition behind this is that the PAYW strategy can achieve greater market penetration compared to the FP strategy. Network externalities expand the advantages of market penetration in a competitive environment, so the competitive sellers are more motivated to choose the PAYW strategy. Our findings can explain some real-world practices, as

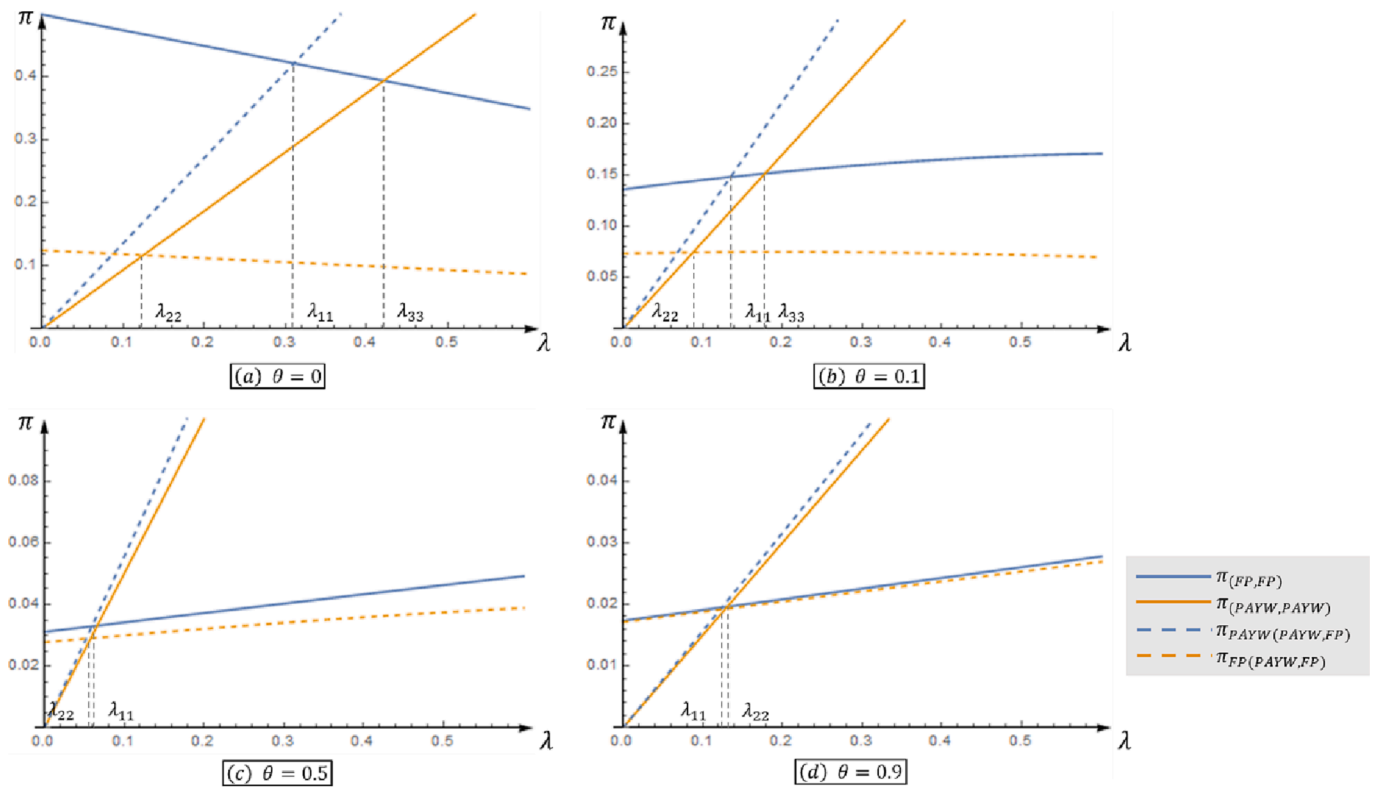


Fig. 9. Sellers' Profits under different θ .

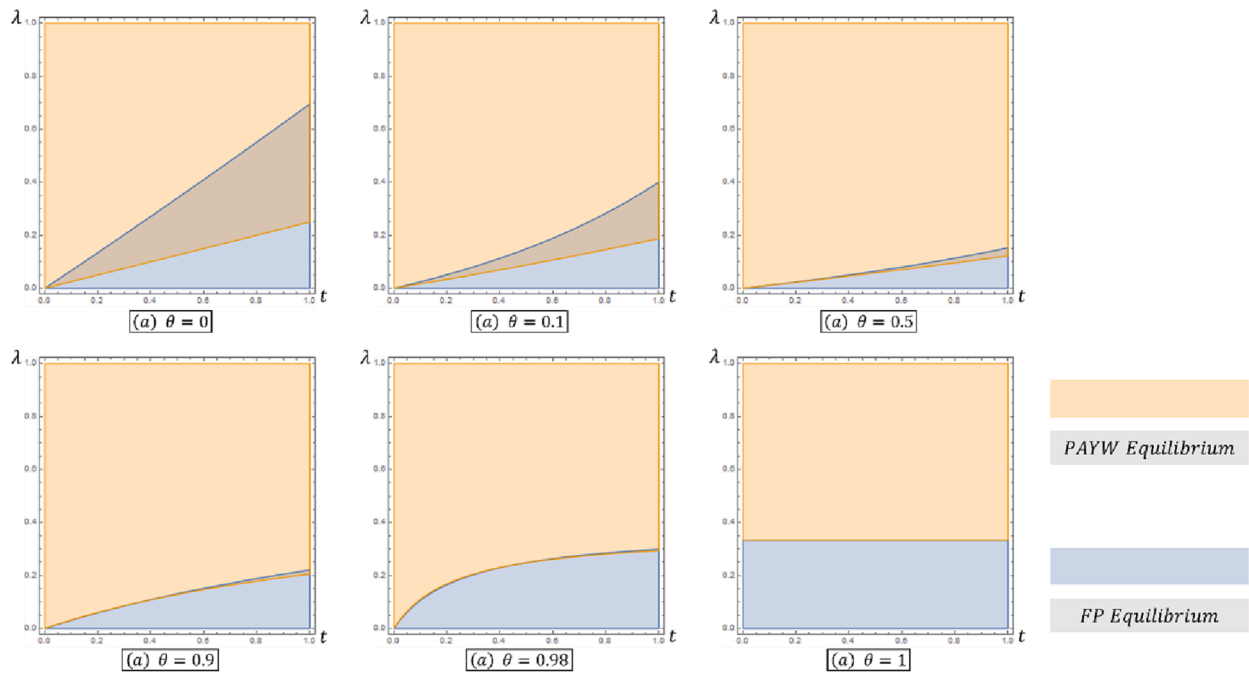


Fig. 10. The likelihood of PAYW equilibrium and FP equilibrium change with θ .

some platforms add social functions (such as comments in a music platform and a bullet screen in a video platform) when selling content products. Furthermore, one can see on the right side of Fig. 5 that $\lambda'_2 < 0$ when $t < \gamma$. This indicates that the PAYW equilibrium always exists no matter the value of consumers fairness ideal. Fig. 6 shows this special situation.

We also investigate the comparative statics of λ_i and λ'_i (where $i = 1, 2, 3$).

It can be seen that λ_i and λ'_i increase with t and μ . This shows that an increase in consumer fairness concern level will make non-Pareto-optimal situations more likely to occur, while a higher competition level will make Pareto-optimal situations more likely to occur, which is not easy to explain intuitively. However, social welfare will always increase with the level of competition, which can be easily proved by combining Tables 3 and 4.

5.3. Heterogenous consumers

In this section, we show that our results in the previous sections qualitatively hold even if consumers are heterogenous in the product value so that the market cannot be fully covered. For simplicity, we assume that there are two types of consumers in the market according to their reservation value: high-type consumers have relatively large reservation value r_H whereas low-type consumers have reservation value $r_L < r_H$. We assume that r_H is large enough so that this type of consumers will always buy the product regardless of their location on Hotelling line. However, for the low-type consumers, we assume that r_L is not large enough so that part of the consumers will leave the market, thus the market is not fully covered. Let θ denote the proportion of low-type consumers and the proportion of high-type consumers is $1 - \theta$. Fig. 7 shows market coverage under the two different types of consumers. With other settings remaining the same, e.g., consumer's utility and seller's profit function, we analyze the equilibrium under this setting following the same logic in section 4 and summarize our findings in the following propositions.

Proposition 5. *When θ is relatively small, the equilibrium outcomes are the same as that stated in Proposition 1 (Fig. 8. (a)). In this case, multiple equilibria and Prisoner's dilemma still exists. Otherwise, asymmetric equilibrium will occur when θ is sufficiently large. That is, one seller chooses PAYW strategy and the other chooses FP strategy to become equilibrium (Fig. 8. (b)). In particular, when $\theta = 1$, the seller's profit is only related to its own pricing strategy and has nothing to do with another seller's pricing strategy.*

Proposition 5 offers three interesting insights about PAYW strategy when consumers are heterogeneous. First of all, all insights from Proposition 1 still qualitatively hold when the proportion of low-type consumers is relatively small. Secondly, when facing a high proportion of low-type consumers, the asymmetric equilibrium (PAYW, FP) can arise. This is a new finding different from the previous analysis. Unlike the asymmetric equilibrium obtained by Chao et al. (2019), in which they consider the impact of external reference price on fair price. We focus on the impact of the existence of heterogeneous consumers on the equilibrium solution. In particular, when there are no high-type consumers in the market, the seller's decision is in fact independent of others. To understand the intuition behind this result, we investigate the impact of θ on each pricing strategy and equilibrium.

Proposition 6. *The impact of θ on the likelihood of the emergence of PAYW equilibrium is not monotone. Specifically, there exists a threshold $\theta_1^*(\mu, \beta, t, r_H, r_L)$. When $\theta < \theta_1^*(\mu, \beta, t, r_H, r_L)$, the likelihood of the emergence of PAYW equilibrium increases with θ . While when $\theta \geq \theta_1^*(\mu, \beta, t, r_H, r_L)$, the likelihood of the emergence of PAYW equilibrium decreases with θ .*

Intuitively, the participation of low-type consumers will inevitably have a negative impact on the profits of PAYW seller and FP seller. However, for PAYW seller, the existence of low-type consumers will not affect the payment of high-type consumers. Thus, the marginal effect of θ on FP seller is greater than that on PAYW seller when θ is not large. At this time, the likelihood of the emergence of PAYW equilibrium increases with θ , while the likelihood of the emergence of FP equilibrium decreases with θ . It can be explained in the similar way when $\theta \geq \theta_1^*(\mu, \beta, t, r_H, r_L)$. This nonmonotonic effect of θ on PAYW strategy is different from the findings obtained in previous literature. For example, Chen et al. (2017) gave such a conclusion in which more low-type consumers can make sellers more motivated to adopt PAYW strategy. He used a trapezoid distribution function to represent the consumer's reservation prices and made such a conclusion under monopoly setting. The different modeling of consumer heterogeneity and the competitive environment explain the contrast findings. These new insights of our

research expand the content of this research field on the basis of previous studies.

6. Conclusion

The PAYW strategy has become a popular pricing strategy for content products in online marketplaces. Although PAYW has attracted considerable attention in the literature, how competition affects the value of PAYW deserves further research. To fill this gap, we build an analytical model to investigate the advantage of PAYW by comparing PAYW with the traditional fixed pricing strategy in a duopoly setting. We incorporate behavioral factors such as fairness concern and reciprocity, which empirical evidence has shown to be the most significant driving factors of PAYW. We analyze the consumers' choices, the sellers' optimal pricing decisions, and the equilibrium pricing strategies. We also extend our results by considering the social welfare and network externality.

Our findings are as follows. First, only symmetric equilibria can exist in our model. That is, both sellers choose the PAYW strategy when facing a higher consumer fairness ideal; otherwise, both choose the FP strategy. We also found that the competition intensity promotes the application of PAYW. Second, we found that the equilibrium may not be unique for a medium consumer's fairness ideal level, which contrasts sharply with prior literature. Moreover, a prisoner's dilemma situation may arise where both sellers adopt the PAYW strategy. However, the prisoner's dilemma situation is optimal for social welfare, and both sellers' profits can be Pareto-improved. Third, the social welfare may not be optimal under the seller's Pareto-optimal equilibrium, and the consumer fairness ideal has a nonmonotonic effect on the social welfare. In particular, a medium fairness ideal level results in the lowest social welfare. Fourth, the existence of network externality will make sellers more likely to choose the PAYW strategy in a competitive market. Finally, we extend our results to the setting with heterogeneous consumers and find that our results still qualitatively hold. Interestingly, we also find when low-value consumers are sufficiently large, asymmetric equilibrium (PAYW, FP) can arise.

CRedit authorship contribution statement

Hongyu Hou: Conceptualization, Methodology, Writing – original draft, Writing – review & editing, Formal analysis. **Feng Wu:** Conceptualization, Methodology, Funding acquisition. **Xiangyin Kong:** Supervision, Methodology, Writing – review & editing, Validation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Notation and proofs

Proof of Lemma 1

Considering the situation when $p_{PAYW} > \lambda(r - tx)$, the utility of consumer buying from seller A is given by

$$u_{PAYW} = r - p_{PAYW} + \alpha p_{PAYW} - \beta[p_{PAYW} - \lambda(r - tx)] - tx$$

Otherwise, the utility of consumer buying from seller A is given by

$$u_{PAYW} = r - p_{PAYW} + \alpha p_{PAYW} - \mu[\lambda(r - tx) - p_{PAYW}] - tx$$

Thus, the marginal utility of consumer when they choose to pay p_{PAYW} under PAYW pricing strategy is given by

$$\frac{du_{PAYW}}{dp_{PAYW}} = \begin{cases} -1 + \alpha - \beta, & \text{if } p_{PAYW} \geq \lambda(r - tx) \\ -1 + \alpha + \mu, & \text{if } p_{PAYW} < \lambda(r - tx) \end{cases}$$

When $\alpha + \mu < 1$, the partial derivative of consumer utility with respect to the payment amount is less than zero, which means that all consumers will not pay any amount. The positive utility brought about by consumers paying a certain price is much smaller than the negative utility brought about by price payment. When $\alpha + \mu \geq 1$, The partial derivative of consumer utility for the payment amount is greater than zero only when $p_{PAYW} < \lambda(r - tx)$, and less than zero when $p_{PAYW} \geq \lambda(r - tx)$. Therefore, consumers will pay the price only when $p_{PAYW} < \lambda(r - tx)$, and the price paid for seller A is the fair price $\lambda(r - tx)$. Considering the assumption that seller A and seller B are symmetric, the same analytical logic can be applied to seller B. Therefore, consumers will pay the price for seller B only when $p_{PAYW} < \lambda(r - tx)$, and the price is the fair price $\lambda[r - t(1 - x)]$.

Proof of Lemma 2

When the sellers adopt FP, two cases arise. Firstly, we analyze the case where the fixed price makes consumers suffer from disadvantageous unfairness. That means, the conditions of this case mathematically are $p_A - \lambda(r - tx) \geq 0$ and $p_B - \lambda[r - t(1 - x)] \geq 0$. In this case, the consumer's utility of getting the content product from seller A is $r - p_A - \beta[p_A - \lambda(r - tx)] - tx$, and from seller B is $r - p_B - \beta[p_B - \lambda[r - t(1 - x)]] - t(1 - x)$. Equating these two formulations gives us the position of the marginal consumer x^* . That means a consumer will purchase the content product from seller A if and only if $x \leq x^* = \frac{t(1+\beta\lambda)+(1+\beta)(p_B-p_A)}{2t(1+\beta\lambda)}$. Thus, the profit of seller A can be expressed as $\pi_A = \frac{t(1+\beta\lambda)+(1+\beta)(p_B-p_A)}{2t(1+\beta\lambda)} \times p_A - c$, and the profit of seller B can be expressed as $\pi_B = \left(1 - \frac{t(1+\beta\lambda)+(1+\beta)(p_B-p_A)}{2t(1+\beta\lambda)}\right) \times p_B - c$. By setting $\frac{d\pi_A}{dp_A} = 0$ and noting the sellers' symmetry, we can easily get that the optimal price of each seller is $\frac{t(1+\beta\lambda)}{1+\beta}$ and the demand of each seller is $\frac{1}{2}$. The profit for each seller under (FP, FP) in this case is $\frac{t(1+\beta\lambda)}{2(1+\beta)} - c$.

Secondly, we analyze the case where the fixed price makes consumers suffer from advantageous unfairness. That means, the conditions of this case mathematically are $p_A - \lambda(r - tx) \leq 0$ and $p_B - \lambda[r - t(1 - x)] \leq 0$. In this case, the consumer's utility when getting the content product from seller A is $r - p_A - \mu[\lambda(r - tx) - p_A] - tx$, and from seller B is $r - p_B - \mu[\lambda[r - t(1 - x)] - p_B] - t(1 - x)$. Similar to the above analysis, we can get that the optimal price of each seller is $\frac{t(1-\mu\lambda)}{1-\mu}$ and the demand of each seller is $x^* = \frac{1}{2}$. The profit for each seller under (FP, FP) in this case is $\frac{t(1-\mu\lambda)}{2(1-\mu)} - c$.

Due to the symmetry of the sellers, the case that one seller makes consumers suffer from advantageous unfairness and another seller makes consumers suffer from disadvantageous unfairness will not occur. Next, we justify which case will be adopted by the sellers by comparing the profits under the above two cases. Since $0 < \lambda < 1$, we can get $\frac{t(1-\mu\lambda)}{2(1-\mu)} > \frac{t}{2} > \frac{t(1+\beta\lambda)}{2(1+\beta)}$. Therefore, the seller's optimal price occurs when the consumer suffers from advantageous unfairness. The optimal price for each seller is $p_A^* = p_B^* = \frac{t(1-\mu\lambda)}{1-\mu}$, and the profit of each seller is $\pi_A^* = \pi_B^* = \frac{t(1-\mu\lambda)}{2(1-\mu)} - c$.

Proof of Lemma 3

According to the analysis of Lemma 1, consumers will pay the fair price for content products under the PAYW strategy. That means $p_{A_PAYW}^* = \lambda(r - tx)$ and $p_{B_PAYW}^* = \lambda[r - t(1 - x)]$ if we rule out free-riding consumers. It is worth mentioning that at this time, consumers will not suffer from the negative effects of either advantageous unfairness or disadvantageous unfairness in this situation. By the sellers' symmetry, we can get the demand of each seller is $x^* = \frac{1}{2}$. Analogous to the previous analysis, the fair price of consumers is $\lambda(r - tx)$ when buying the product from seller A and $\lambda[r - t(1 - x)]$ when buying from seller B. The profits of seller A and seller B can then be expressed as $\pi_A = \pi_B = \int_0^{\frac{1}{2}} \lambda(r - tx) dx - c$. The profit for each seller under (PAYW, PAYW) is given by $\pi_A = \pi_B = \frac{\lambda}{2} (r - \frac{t}{4}) - c$.

Proof of Lemma 4

Assume that seller A adopts the PAYW strategy, and seller B adopts the FP strategy. Before continuing the analysis, we need to explain which consumers buy the product from seller A and which from seller B. Following the market segmentation of Chen et al. (2017) which is stated in their Proof of Proposition 6, we give the proof logic of Lemma 4. Firstly, we need to calculate the maximum willingness to pay of consumers for seller A given seller B's price. Secondly, we will find the location of marginal consumers (indifferent point of buying from seller A and seller B) based on the result of the first step. Thirdly, we need to give the profit functions of the two sellers. And then, we can find the optimal fixed price of seller B by solving the profit maximization problem. Finally, we get the profits of two sellers under this sub-game. Since seller A does not make pricing decision, seller B's pricing decision will lead to different results. According to the fixed price set by seller B, we divide the proof into two cases.

Firstly, we analyze the case where the fixed price makes marginal consumers suffer from disadvantageous unfairness. In this case, the utility of consumer buying from seller B is expressed as $u_B = r - p_B - \beta[p_B - \lambda[r - t(1 - x)]] - t(1 - x)$. Denote the consumer's maximum willingness to pay to seller A given seller B's price as p_A . Then, the utility of consumer buying from seller A is expressed as $u_A = r - p_A + \alpha p_A - \beta[p_A - \lambda(r - tx)] - tx$. Letting $u_A \geq u_B$, we can calculate a consumer's maximum willingness to pay for seller A's content product given p_B is $p_A \leq \frac{1+\beta}{1+\beta-\alpha} p_B + \frac{t(1-2x)(1+\beta\lambda)}{1+\beta-\alpha}$, where p_B is the fixed price set by seller B. Then following Chen et al. (2017), we reasonably assume that consumers will get the content product from seller A if and only if $p_A > 0$. Because in this case, consumers who go to seller A to get this content product can get higher utility than by going to seller B. Otherwise, consumers will go to seller B to buy the product at the fixed price p_B . By setting $\frac{1+\beta}{1+\beta-\alpha} p_B + \frac{t(1-2x)(1+\beta\lambda)}{1+\beta-\alpha} = 0$, we can get the location of the marginal consumers $x^* = \frac{1+\beta}{2t(1+\beta\lambda)} p_B + \frac{1}{2}$. Then, the demand of seller A is $\frac{1+\beta}{2t(1+\beta\lambda)} p_B + \frac{1}{2}$, and the demand of seller B is $\frac{1}{2} - \frac{1+\beta}{2t(1+\beta\lambda)} p_B$. According to the above analysis, the seller B's profits are given by $\pi_B = (1 - x^*) \times p_B - c = \left[\frac{1}{2} - \frac{1+\beta}{2t(1+\beta\lambda)} p_B\right] \times p_B - c$. According to the principle of profit maximization, we find seller B's the optimal price $p_B = \frac{t(1+\beta\lambda)}{2(1+\beta)}$ by setting $\frac{d\pi_B}{dp_B} = 0$. Then, we can find that $x^* = \frac{1+\beta}{2t(1+\beta\lambda)} p_B + \frac{1}{2} = \frac{3}{4}$. The profit seller B is given by $\pi_B = \frac{t(1+\beta\lambda)}{8(1+\beta)} - c$.

Secondly, we analyze the case where the fixed price makes marginal consumers suffer from advantageous unfairness. In this case, the utility of consumer buying from seller B is expressed as $u_B = r - p_B - \mu\{\lambda[r - t(1-x)] - p_B\} - t(1-x)$. Similar to the analysis in the above part, the utility of consumer buying from seller A is expressed as $u_A = r - p_A + \alpha p_A - \mu[\lambda(r - tx) - p_A] - tx$. Letting $u_A \geq u_B$, a consumer's maximum willingness to pay for seller A's content product given p_B is $p_A \leq \frac{1-\mu}{1-\mu-\alpha}p_B + \frac{t(1-2x)(1-\mu\lambda)}{1-\mu-\alpha}$. By setting $\frac{1-\mu}{2t(1-\mu\lambda)}p_B + \frac{1}{2} = 0$, we can get the location of the marginal consumers $x^* = \frac{1-\mu}{2t(1-\mu\lambda)}p_B + \frac{1}{2}$. Then, the demand of seller A is $\frac{1-\mu}{2t(1-\mu\lambda)}p_B + \frac{1}{2}$, and the demand of seller B is $\frac{1}{2} - \frac{1-\mu}{2t(1-\mu\lambda)}p_B$. Similarly, according to the above analytical logic, we can get the optimal price of seller B is $\frac{t(1-\mu\lambda)}{2(1-\mu)}$, and the position of the marginal consumer is $x^* = \frac{3}{4}$. The seller B's profit is $\pi_B = \frac{t(1-\mu\lambda)}{8(1-\mu)} - c$.

Next, we justify which fixed price will be adopted by seller B through comparing the profits under the above two cases. Since $0 < \lambda < 1$, we can easily prove that $\frac{t(1-\mu\lambda)}{8(1-\mu)} > \frac{t}{8} > \frac{t(1+\beta\lambda)}{8(1+\beta)}$. Thus, the optimal fixed price of seller B is $p_B^* = \frac{t(1-\mu\lambda)}{2(1-\mu)}$, and the optimal profit of seller B is $\pi_B^* = \frac{t(1-\mu\lambda)}{8(1-\mu)} - c$. For those consumers who get the content product from seller A, they will pay the fair price expressed as $\lambda(r - tx)$ according to the analysis of Lemma 1. Thus, the profit of seller A is given by $\pi_A = \int_0^{x^*} \lambda(r - tx)dx - c = \frac{3}{4}\lambda(r - \frac{3}{8}t) - c$.

Proof of Proposition 1

In equilibrium, neither seller has an incentive to change their pricing strategy. As Table 4 shows, if both sellers choosing FP is an equilibrium, we must have $\frac{t(1-\mu\lambda)}{2(1-\mu)} - c \geq \frac{3}{4}\lambda(r - \frac{3}{8}t) - c$. From this, we can get $\lambda \leq \frac{16t}{24r-24r\mu+25t\mu-9t} = \lambda_1$. If both sellers choosing PAYW is an equilibrium, $\frac{1}{2}(r - \frac{t}{4}) - c \geq \frac{t(1-\mu\lambda)}{8(1-\mu)} - c$ is required, leading to $\lambda \geq \frac{t}{4r-4r\mu+2t\mu-t} = \lambda_2$. If both sellers choosing PAYW is Pareto-optimal, $\frac{1}{2}(r - \frac{t}{4}) - c \geq \frac{t(1-\mu\lambda)}{2(1-\mu)} - c$ is required. That means that only when $\lambda \geq \frac{4t}{4r-4r\mu+5t\mu-t} = \lambda_3$ will both sellers choosing PAYW be a Pareto-optimal equilibrium. We can easily prove that $0 < \lambda_1 < 1$, $0 < \lambda_2 < 1$, $0 < \lambda_3 < 1$, and $\lambda_3 > \lambda_1 > \lambda_2$ because the r is large enough to cover the full market.

Proof of Proposition 2

Since there are only two equilibrium outcomes, we analyze the consumer surplus and social welfare under these two equilibria. According to Table 5, if the consumer surplus under the (PAYW, PAYW) equilibrium is higher than that under the (FP, FP) equilibrium, $\frac{(4r-t)(1-\lambda+\alpha\lambda)}{8} > \frac{(4r-5t)(1-\mu\lambda)}{8}$ is required. From this, we can prove $\lambda > \frac{4t}{4r(1-\mu-\alpha)+5t\mu-t+\alpha t} = \lambda_4$. It is obvious that $\lambda_4 < 0$ because $\alpha + \mu > 1$.

If the social welfare under the (PAYW, PAYW) equilibrium is higher than that under the (FP, FP) equilibrium, we must have $\frac{(4r-t)(1-\lambda+\alpha\lambda)}{8} + \frac{1}{2}(r - \frac{t}{4}) - c > \frac{(4r-5t)(1-\mu\lambda)}{8} + \frac{t(1-\mu\lambda)}{2(1-\mu)} - c$. From this, we get $\lambda > \frac{4t}{4r(1-\mu)(1+\frac{\alpha}{\mu})+5t\mu-t+\alpha t\frac{\alpha}{\mu}} = \lambda_5$. We can easily prove that $\lambda_5 < \lambda_3$.

Proof of Proposition 3

Given the expression of λ_1 , λ_2 , and λ_5 in the above proofs, we can prove that $\alpha < \alpha_1 = \frac{8r\mu-5t\mu}{4(4r+t)}$ when $\lambda_5 > \lambda_1$. Similarly, if $\lambda_5 < \lambda_2$, we can get $\alpha > \alpha_2 = 3\mu$. Therefore, we get the following results. In the case of $\alpha < \alpha_1$, we get $\lambda_3 > \lambda_5 > \lambda_1 > \lambda_2$. In the case of $\alpha_1 \leq \alpha \leq \alpha_2$, we get $\lambda_3 > \lambda_1 \geq \lambda_5 \geq \lambda_2$. In the case of $\alpha > \alpha_2$, we get $\lambda_3 > \lambda_1 > \lambda_2 > \lambda_5$.

Proof of Proposition 4

If both sellers choosing FP is an equilibrium, then $\frac{t(1-\mu\lambda)}{2(1-\mu)} - c \geq \frac{3}{4}\lambda(r - \frac{3}{8}t + \frac{3}{4}\gamma) - c$ is required, leading to $\lambda \leq \frac{16t}{24r-24r\mu+25t\mu-9t+18\gamma\mu} = \lambda'_1$. If both sellers choosing PAYW is an equilibrium, then $\frac{1}{2}(r - \frac{t}{4} + \frac{1}{2}\gamma) - c \geq \frac{t(1-\mu\lambda)}{8(1-\mu)} - c$ is required, and $\lambda \geq \frac{t-\gamma}{4r-4r\mu+2t\mu-t+3\gamma-3\gamma\mu} = \lambda'_2$. It can be easily proved that $\lambda'_2 < \lambda'_1$. If both sellers choosing PAYW is Pareto-optimal, then $\frac{1}{2}(r - \frac{t}{4} + \frac{1}{2}\gamma) - c \geq \frac{t(1-\mu\lambda)}{2(1-\mu)} - c$ is required. That means the only when $\lambda \geq \frac{4t}{4r-4r\mu+5t\mu-t+2\gamma-2\gamma\mu} = \lambda'_3$ will both sellers choosing PAYW will be a Pareto-optimal equilibrium. From these, it can be proved that $\lambda'_3 > \lambda'_1 > \lambda'_2$, $\frac{d\lambda'_3}{d\gamma} > 0$, and $\frac{d\lambda'_3}{d\gamma} < 0$.

Proof of Proposition 5

1. When the sellers adopt FP strategy, we discuss the marginal consumer's position of low-type consumers firstly. Supposing that the marginal consumer is affected by advantageous unfairness, $p_A < p_{fair}$. Thus, the consumer's utility function is $u_A = r_L - p_A - \mu[\lambda(r_L - tx) - p_A] - tx$. So, the position of the marginal consumer is $x_A^* = \frac{r_L(1-\mu\lambda) - (1-\mu)p_A}{t(1-\mu\lambda)}$. The fair price perceived by the marginal consumer is $p_{fair} = \lambda(r_L - tx_A^*) = \frac{\lambda(1-\mu)p_A}{t(1-\mu\lambda)}$. It is easy to prove that $\frac{\lambda(1-\mu)p_A}{t(1-\mu\lambda)} < p_A$. There's a paradox here. So, the marginal low-type consumer is affected by the disadvantageous unfairness. For seller A, the marginal high-type consumer's position is $x_H^* = \frac{t(1-\mu\lambda) + (1-\mu)(p_B - p_A)}{2t(1-\mu\lambda)}$ (as shown in Lemma 2), and the marginal low-type consumer's position is $x_A^* = \frac{r_L(1+\beta\lambda) - (1+\beta)p_A}{t(1+\beta\lambda)}$. Therefore, the profit of seller A can be expressed as $\pi_A = (1-\theta)\frac{t(1-\mu\lambda) + (1-\mu)(p_B - p_A)}{2t(1-\mu\lambda)} \times p_A + \theta\frac{r_L(1+\beta\lambda) - (1+\beta)p_A}{t(1+\beta\lambda)} \times p_A - c$. By setting $\frac{dp_A}{dp_A} = 0$ and noting the sellers' symmetry, we can easily get the optimal price of each seller is $p_A^* = p_B^* = \frac{(1-\theta)(1-\mu\lambda)(1+\beta\lambda)t + 2\theta r_L(1-\mu\lambda)(1+\beta\lambda)}{(1-\theta)(1-\mu)(1+\beta\lambda) + 4\theta(1+\beta)(1-\mu\lambda)}$ and the profit of each seller is $\pi_A^* = \pi_B^* = \frac{(1+\beta\lambda)(1-\mu\lambda)[(1-\theta)(1-\mu)(1+\beta\lambda) + 2\theta(1+\beta)(1-\mu\lambda)] + 2\theta r_L + (1-\theta)t^2}{2t[(1-\theta)(1-\mu)(1+\beta\lambda) + 4\theta(1+\beta)(1-\mu\lambda)]} - c$.

2. When the sellers adopt PAYW strategy, we discuss the marginal consumer's position of low-type consumers firstly. By setting $u_{AL} = r_L - p_{AL} + \alpha p_{AL} - \beta \max\{p_{AL} - \lambda(r_L - tx), 0\} - \mu \max\{\lambda(r_L - tx) - p_{AL}, 0\} - tx = 0$, we can get the marginal low-type consumer's position of seller A is $x_{AL}^* = \frac{r_L}{t}$. Thus, the seller's profit is given by $\pi_A = \pi_B = (1-\theta)\int_0^{\frac{r_L}{t}} \lambda(r_H - tx)dx + \theta\int_0^{\frac{r_L}{t}} \lambda(r_L - tx)dx - c = \frac{(1-\theta)\lambda r_H}{2} - \frac{(1-\theta)\lambda t}{8} + \frac{\theta\lambda r_L^2}{2t} - c$.

3. Assume that seller A adopts the PAYW strategy, and seller B adopts the FP strategy. Combined with the analysis in Lemma 4, we get the marginal high-type consumer's position is $x_H^* = \frac{1-\mu}{2t(1-\mu\lambda)}p_B + \frac{1}{2}$. The marginal low-type consumer's position of PAYW seller is $x_{AL}^* = \frac{r_L}{t}$ and of FP seller is $x_{BL}^* = 1 - [\frac{r_L}{t} - \frac{1+\beta}{t(1+\beta\lambda)}p_B]$. The profit of FP seller can be express as $\pi_B = (1-\theta)[\frac{1}{2} - \frac{1-\mu}{2t(1-\mu\lambda)}p_B] \times p_B + \theta[\frac{r_L}{t} - \frac{1+\beta}{t(1+\beta\lambda)}p_B] \times p_B - c$. Thus, we can get the optimal fixed price set by seller B is $p_B^* = \frac{[2\theta r_L + (1-\theta)t](1-\mu\lambda)(1+\beta\lambda)}{2[(1-\theta)(1-\mu)(1+\beta\lambda) + 2\theta(1+\beta)(1-\mu\lambda)]}$. The profit of PAYW seller is $\pi_A = (1-\theta)\int_0^{x_H^*} \lambda(r_H - tx)dx + \theta\int_0^{\frac{r_L}{t}} \lambda(r_L - tx)dx - c = \frac{(1-\theta)\lambda(MY + 2tA)[8Ar_H - (MY + 2tA)]}{32tA^2} + \frac{\theta\lambda r_L^2}{2t} - c$. The profit of FP seller is $\pi_B = (1-\theta)(1 - x_{BL}^*) \times p_B + \theta(1 - x_{BL}^*) \times p_B - c = \frac{[(1-\theta)(2tA - MY) + 2\theta(2r_LA - MX)]M(1-\mu\lambda)(1+\beta\lambda)}{8tA^2} - c$. Note that for convenience, we make $X = (1+\beta)(1-\mu\lambda)$, $Y = (1-\mu)(1+\beta\lambda)$, $M = 2\theta r_L + (1-\theta)t$ and $A = [(1-\theta)(1-\mu)(1+\beta\lambda) + 2\theta(1+\beta)(1-\mu\lambda)]$.

4. In equilibrium, neither seller has an incentive to deviate from their own pricing strategy. If both sellers choose FP strategy is an equilibrium,

$\frac{(1+\beta\lambda)(1-\mu\lambda)AM^2}{2t(A+2\theta X)^2} \geq \frac{(1-\theta)\lambda(MY+2tA)[4Ar_H-(MY+2tA)]}{16tA^2} + \frac{\theta\lambda r_L^2}{2t}$ is required. It can be solved that $\lambda \leq \lambda_{11}(\theta, \mu, \beta, t, r_H, r_L)$. If both sellers choose PAYW strategy is an equilibrium, $\frac{(1-\theta)\lambda r_H}{2} - \frac{(1-\theta)\lambda t}{8} + \frac{\theta\lambda r_L^2}{2t} \geq \frac{[(1-\theta)(2tA-MY)+2\theta(2r_LA-MX)]M(1-\mu\lambda)(1+\beta\lambda)}{8tA^2}$ is required. It can be solved that $\lambda \geq \lambda_{22}(\theta, \mu, \beta, t, r_H, r_L)$. If both sellers choose PAYW strategy is Pareto-optimal, $\frac{(1-\theta)\lambda r_H}{2} - \frac{(1-\theta)\lambda t}{8} + \frac{\theta\lambda r_L^2}{2t} \geq \frac{(1+\beta\lambda)(1-\mu\lambda)AM^2}{2t(A+2\theta X)^2}$ is required. It can be solved that $\lambda \geq \lambda_{33}(\theta, \mu, \beta, t, r_H, r_L)$.

Considering that it is difficult to find the analytical solution, we draw figures for analysis.

We can get some results from the Fig. 9. When θ is not relatively large, we can see $\lambda_{33} > \lambda_{11} > \lambda_{22}$, which is similar to the equilibrium structure we got in Proposition 1. When θ is relatively large, as shown in Fig. 10. (d), we can see $\lambda_{11} < \lambda_{22}$. This means that when the consumer's fairness ideal satisfies $\lambda_{11} < \lambda < \lambda_{22}$, the market is in asymmetric equilibrium. Because at this time, both the sellers under the (FP, FP) strategy and the (PAYW, PAYW) strategy have incentives to change their pricing strategies.

Consider a special case, that is, $\theta = 1$. We can calculate the seller's profit under (FP, FP) strategy is $\pi_{(FP,FP)} = \frac{r_L^2(1+\beta\lambda)}{4t(1+\beta)} - c$, the seller's profit under (PAYW, PAYW) strategy is $\pi_{(PAYW,PAYW)} = \frac{\lambda r_L^2}{2t} - c$. The PAYW seller's profit under (PAYW, FP) strategy is $\pi_{PAYW(PAYW,FP)} = \frac{\lambda r_L^2}{2t} - c$ and the FP seller's profit under (PAYW, FP) strategy is $\pi_{FP(PAYW,FP)} = \frac{r_L^2(1+\beta\lambda)}{4t(1+\beta)} - c$. Interestingly, we find that $\pi_{(FP,FP)} = \pi_{FP(PAYW,FP)}$ and $\pi_{(PAYW,PAYW)} = \pi_{PAYW(PAYW,FP)}$. This means that there is no competition in the market. And the seller's profit is only related to its own pricing strategy choice.

Proof of Proposition 6

In this part, we consider to analyze the changing trend of $\lambda_{11}(\theta, \mu, \beta, t, r_H, r_L)$ and $\lambda_{22}(\theta, \mu, \beta, t, r_H, r_L)$ with θ . Because it is difficult to give specific expressions of $\lambda_{11}(\theta, \mu, \beta, t, r_H, r_L)$ and $\lambda_{22}(\theta, \mu, \beta, t, r_H, r_L)$ in Proposition 5. We consider using figures to give a clear explanation. We can see from Fig. 10 that the yellow area represents the scope of the PAYW equilibrium, and the blue area represents the scope of the FP equilibrium. The scope of PAYW equilibrium increases at first and then decreases with θ . Thus, we can get $\theta_1^*(\mu, \beta, t, r_H, r_L)$ by making $\frac{d\lambda_{11}(\theta, \mu, \beta, t, r_H, r_L)}{d\theta} = 0$. While the scope of FP equilibrium decreases at first and then increases with θ . Similarly, we can get a $\theta_2^*(\mu, \beta, t, r_H, r_L)$. Summing up the above findings, we get Proposition 6.

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